

Data - Vocabulary

database: data, set of relations/populations/samples/datasets/tables
dataset: data, population, database, relation, sample, table
variable: column name, feature, attribute, dimension, field
observation #: record/row # value: datum, element, cell, item
vector: column, array, distribution observation: record, datum, tuple
data(pl.) / datum(singular) vector, datapoint, row, instance, e.g.
variable: characteristic / quantity of interest w/ diff. values
decision var: value directly determined by decision-maker
random var: " not directly determined by decision-maker
domain: set of poss. values for a var population: all obs. of
obs: set of values corresponding to a set of var interest
sample: subset of all obs. of interest
categorical data: data where numerical operations CAN'T be performed
numerical data: " " " CAN be performed
(cross-sectional data: order of observations NOT meaningful
time-series data: order IS meaningful
experimental study: var of interest identified, other vars controlled
observational study: vars are not controlled (a survey)

1 var: re: variation normalization: any units
range: largest - smallest $\Rightarrow$ standard
variance / standard deviation z-score: $z = \frac{x - \bar{x}}{s}$ $\bar{x}$: sample mean
 σ^2 / σ_x^2 : pop. var σ / σ_x : pop. std dev s : sample std. dev
 s^2 / s_x^2 : sample var s / s_x : sample std. dev $\approx 68\%$ $-1 < z < 1$
 $\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$; $\sigma = \sqrt{\sigma^2}$ $s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$; $s = \sqrt{s^2} \approx 95\%$ $-2 < z < 2$
pth percentile: value of var @ specified % of almost all:
observations below that value (x p% of obs. < pth)
Find pth percentile:
- arrange vals in ascending order
- compute $k = (n+1) \cdot p/100$ $n = \#$ of values
- split k into integer i and decimal d components below Q_1
- interpolate b/t ith and i+1th positions' values OR 1.5(IQR)
- $m = \text{val @ (i+1)th pos.} - \text{val @ ith pos.}$ addtl: above Q_3
- $t = m \times d$ sigma process: limits
- pth percentile = value @ ith pos + t for a normal process
quartile / inter-quartile range 99.9996% values b/t -6 < z < 6
- Q_1 (25th percentile) Q_2 (50th percentile / median) 0 = low
- Q_3 (75th percentile) IQR = $Q_3 - Q_1$ high neg | high
covariance / correlation correlation $\Rightarrow$ # from -1 to 1
 σ_{xy} : pop. covariance b/t x & y r = correlation coeff.
 s_{xy} : sample covariance b/t x & y σ_x, σ_y = pop std dev for x & y
 s_x, s_y = sample std. dev x, y
 $\sigma_{xy} = \frac{\sum (x_i - \mu_x)(y_i - \mu_y)}{N}$
 $s_{xy} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{n-1}$
 $r = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$ $-1 \leq r \leq 1$
 $r = \frac{s_{xy}}{s_x s_y}$ descriptive stats
Continued

Descriptive Statistics

Frequency dist: list of # summarizing how often specific values occur in a dataset:
1 categorical var 1 numerical var
- frequency dist. - same 3 as categorical
- rel. frequency dist. - cumulative freq.
- percent frequency dist. - rel. cumulative
- % cumulative
for numerical, can pick:
- # nonoverlapping bins, width of each bin / bin limits
- successive bin size = sum of prev (cum)
1 var: re: location $\bar{x}$ for sample arithmetic mean: $\bar{X}$ for population
geometric mean μ_g : pop $\bar{X}_g$: sample
 $\mu_g = \sqrt[n]{\prod_{i=1}^n x_i} = \sqrt[n]{x_1 x_2 \dots x_n}$
 $\bar{X}_g = \sqrt[n]{\prod_{i=1}^n x_i} = \sqrt[n]{x_1 x_2 \dots x_n}$
median: odd $\Rightarrow$ middle even $\Rightarrow$ mean of 2 middle
mode: most frequent

Data Visualization

Scatterplot, lineplot, barplot
pie chart heatmap box plot
histogram right skew symmetric
scatterplot for correlation
cross-tabulation: summarize large dataset w/ small pivot
table w/ var names & obs. names
aggregate measurement values (columns/rows)
Eg. Yearly Rate of Return
1 rate $\Rightarrow$ growth factor (1 + return) 2 n-th root
3 multiply growths over N years 4 subtract 1

Cluster Analysis

Euclidean (numerical): highly influenced by unit of similarity

$$z\text{-score}(a) = \left[\frac{\text{obs. a var1} - \text{obs. b var1}}{S_{\text{var1},1}} \dots \frac{\text{obs. a vark} - \text{obs. b vark}}{S_{\text{var1},k}} \right]$$

Matching coeff. (BINARY categorical): SIMILARITY
 1 if a == b, 0 else 0 $\Rightarrow a_i = \text{obs. a var } i$
 $(a_1, 0, 0, \dots) + \dots + (a_k, 0, 0, \dots)$
 $k = \# \text{ of variables}$

Jaccard's coeff. (BINARY categorical): SIMILARITY
 frac w/ matching NON-ZERO values (doesn't count matching zero values; avoid mistaking absence)

OMIT MATCHING then same formula as matching

dissimilarity matrix: pairwise dissimilarity

similarity matrix: pairwise similarity

single linkage: similarity of clusters = similarity of most similar pair of obs.

complete linkage: similarity of farthest pair of obs.

group average: mean of all pairs of observations

centroid: similarity of centroids (centroid = mean of obs.)

ward's: minimize resulting $\uparrow$ in dissimilarity within new cluster

incentive: minimize resulting $\uparrow$ in σ^2 across cluster

K-Means Method

1. decide # of clusters

2. randomly choose a centroid / cluster

3. repeat: re-assign obs. to nearest cluster (centroid)

calc. resulting centroid of each cluster

Hierarchical Agglomeration Method (HAM)

1. start w/ each obs. assigned to own cluster

2. iteratively combine 2 clusters most similar

$\rightarrow$ decide how measure similarity b/w obs. and clusters

3. decide # of clusters

dendrogram: visualization of HAM

choose # clusters, select cutline

evals:

mean inter-cluster diss: for each pair, dist b/t centroids (pair of clusters)

mean intra-cluster diss: find dist to centroid (obs. in)

mean mean: mean of intra-cluster dist.

dispersion ratio: $\frac{\text{mean inter}}{\text{mean intra}}$

Association Rules: Antecedent (A) $\Rightarrow$ consequent (C)

confidence: $\frac{\text{supp}(A \& C)}{\text{supp}(A)}$

lift: $\frac{\text{supp}(A \& C)}{\text{supp}(A) \cdot \text{supp}(C)}$

benefit: $\frac{\text{lift}}{\text{lift} - 1} - 1$

lift = $\frac{P(A \& C)}{P(A) \cdot P(C)}$

confidence = $P(A \& C | A)$

Classification

TP (acc1, pred1)

TN (acc0, pred0)

FP (acc0, pred1)

FN (acc1, pred0)

T = # obs. total

TPR (recall)

TNR (specificity)

FPR (class error)

FNR (class error)

ROC (goodness of fit)

pick obs. w/ highest prob

how many acc pos. class

0.5 < AUC < 1.0 $\Rightarrow$ AUC < 0.5 invert model!

Regression

avg. error = $\frac{\sum e_i}{n}$

RMSFE = $\sqrt{\frac{\sum e_i^2}{n}}$

evaluation

train / val / test $\Rightarrow$ reduce risk overfitting

$\Rightarrow$ train model 1/2 $\Rightarrow$ take model w/ highest val

data cleaning:

discard obs. w/ missing vals, outliers, errors

discard var. w/ ...

fill in missing entries w. est. val (imputation)

leave data, use methods that account for

reduce # of vars (non-critical, highly correlated, PCA)

re-express categorical as numerical

re-scale vars / re-express

Classification

pred

1 0 sum

TP FN TP+FN

FP FN FP+TN

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TNR

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ROC

pick obs.

how many acc

0.5 < AUC

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Stochastic Modeling

uncertainty = risk risk analysis = decisions based on uncertain var and outcomes
 numerical categorical
 normal binomial
 uniform poisson
 binomial: n indep. trials
 $\text{prob}(A) = p$ $\text{prob}(B) = 1-p$
 distribution of # trials w/outcome A
 review percentile: pth
 percentile = p% values $\leq$ val
 law of averages:
 $\text{mean}(f(x,y,...z)) \approx f(\text{mean}(x), \dots, \text{mean}(z))$
 monte carlo simulation
 make a decision $\rightarrow$ characterize risk $\rightarrow$ benefit:
 identify outcome
 iterate thru many trials
 assume certainty for this trial
 determine outcome based on certainty for this trial
 track outcome
 tally outcomes w/ descriptive stat

alt. business obj.
 maximize expected benefit: max mean
 minimize variation of benefit: min std.
 maximize ret. to risk: max ratio of mean to std dev.
 maximize worst case benefit: max min
 maximize best case: max max
 minimize loss
 risk: min prob neg benefit
 max downside: max max benefit below low prob.
 max upside: max min benefit above low prob.

Decision var uncertain var output var summary var
 static dynamic dynamic tallied
 each is a single val each is a single val from dist. (recalc. each trial) on dist. of outcome

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Optimization by Linear Programming

Decision var	Objective func.	constraints	non-neg constraint
Indicate domain of variables	max or min express in terms of decision var	equalities/inequalities in terms of decision var	in terms of decision variables

graphing constraints:
 non-neg constraints = axes
 other constraints: each line cuts 2D space into satisfying vs. unsatisfying parts
 feasible region via shading (extreme = vertices)
 objective func. is set of lines where each line is a set of objective func. solutions around fixed value $\rightarrow x_1, x_2$ comb $\Rightarrow$ same fixed value; farther from origin = $\uparrow$ fixed values, highest fixed value line intersects feasible region @ one extreme (optimal)

terms:
 binding constraint: constraint change, sol change
 nonbinding constraint: constraint change, sol no change
 slack in nonbinding: how much $\leq$ constraint can be strengthened before becoming binding = unused resource
 surplus in nonbinding: how much $\geq$ constraint can be relaxed before becoming binding = amt $>$ than some required min level
 infeasibility: no solution (can't satisfy all constraints)
 unboundedness: infinitely large/small w/o constraint violations

Sensitivity Analysis for Optimization Models

How sensitive are decisions to parameters / how sensitive is objective function to the parameters?
 Coefficient Allowable Increase/Decrease
 DECISION VARIABLE! how much can you change (associated with) objective without changing the decision
 how much decision var. coeff needs to $\uparrow/\downarrow$ to force change in decision var. value (NOT how obj. func. val change)

Shadow Price
 CONSTRAINT! $a_1x_1 + a_2x_2 + \dots \leq \text{rhs}$ or $\geq \text{rhs}$ (associated with)
 how much optimal soln would change if constraint were changed / how much obj. func. val @ optimal soln would increase if rhs of constraint $\uparrow$ by 1
 constraint rhs change $\rightarrow$ obj. func. val change, decision var. values change shadow = 0, nonbinding (indicates how obj. func. val could change)

Constraint Allowable Increase/Decrease
 CONSTRAINT! how much can you change a (associated with) constraint w/ a predictable effect on opt soln / how much rhs needs to $\uparrow/\downarrow$ to invalidate shadow price.

Reduced Cost
 DECISION VAR what would it take to decide (associated with) to use one thing over another
 if decision var 0 @ opt. soln, how much would its coeff. need to decrease to cause dec. var. to become non-zero
 opportunity cost (similar to reduced cost) obj. value changes!
 if dec var 0 @ opt. soln, how much would obj. func. value increase if decision var. increased from 0 $\rightarrow$ 1

canonical form:
 $\max c_1x_1 + c_2x_2 + \dots + c_nx_n = \min -c_1x_1 - c_2x_2 - \dots - c_nx_n$
 $a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1 \Rightarrow -a_{11}x_1 - a_{12}x_2 - \dots - a_{1n}x_n \geq -b_1$
 $a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \leq b_1$
 $a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n \geq b_1$

product mix eg: $x_1, x_2 \in \text{continuous}$ (amt of A, amt of B)
 objective: $\min 2x_1 + 3x_2$ (minimize cost)
 constraint:
 $x_1 + x_2 \geq 350$ (total production required)
 $x_1 \geq 125$ (prod A production required) $x_1, x_2 \geq 0$
 $2x_1 + x_2 \leq 600$ (processing time limit)
 transport eg: $x_{AC}, x_{AD}, x_{BC}, x_{BD} \geq 0$, cont.

$\min x_{AC} + x_{AD} + 2x_{BC} + 2x_{BD}$ $x_{AC} + x_{AD} \leq A$ $x_{BC} + x_{BD} = C$
 alt. optimal: make new obj. func., sum of dec. vars = 0 at optimal solution
 keep constraints from original model
 add constraint: optimal obj. func. val stay same
 solve new opt. solution, if pos. $\Rightarrow$ found alt.

Integer LP: convex-hull
 LP relaxation: LP dropping integer req.
 convex hull: smallest convex polygon containing all integer-feasible points (lattice points in/in side polygon); corners are int.
 integer optimum: corner of hull in obj.
 improving direction (for max, $\leq$ LP relax opt.)

LP's continued

Project go/no-go model

binary decision var (switches)

A B Total cap

PV a b *

Year 1 c d *x

$x_1, x_2 \in \text{binary}$

max $ax_1 + bx_2$

$$cx_1 + dx_2 \leq x$$

multiple choice constraint

x_2, x_3, x_4 3 proj, require 1

$$x_2 + x_3 + x_4 = 1$$

can do 1: $x_2 + x_3 + x_4 \leq 1$

(mutually exclusive)

k out of n alt. constraints:

require 2: $x_2 + x_3 + x_4 = 2$

up to 2: $x_2 + x_3 + x_4 \leq 2$

conditional: only if

2 could be ~~built~~ if 3 is built

$$x_2 \leq x_3$$

2 must be iff 3 is built

$$x_2 = x_3$$

Fixed cost model

binary var = whether or not

an activity is undertaken

fixed cost: Setup cost if

an activity is undertaken

variable cost: cost directly

related to prod. quantity

x_1 : amt fuel prod x_4 : if fuel

x_2 : amt solv prod x_5 : if solv

x_3 : amt carpet clean prod x_6 : if carpet

x_1, x_2, x_3 continuous x_4, x_5, x_6 binary

$$\max 40x_1 + 30x_2 + 50x_3 - 200x_4 - 50x_5 - 400x_6$$

$$0.4x_1 + 0.5x_2 + 0.6x_3 \leq 20 \quad \left\{ \begin{array}{l} \text{material} \\ \text{constraint} \end{array} \right.$$

$$0.2x_2 + 0.1x_3 \leq 5$$

$$0.6x_1 + 0.3x_2 + 0.3x_3 \leq 2$$

$$x_1 \leq 50x_4 \quad x_2 \leq 25x_5 \quad x_3 \leq 40x_6$$

generally:

start w/ LP relaxation

add var: binary var as switches

associated w/ processes w/ fixed costs

revised obj. func: attach switches to fixed cost

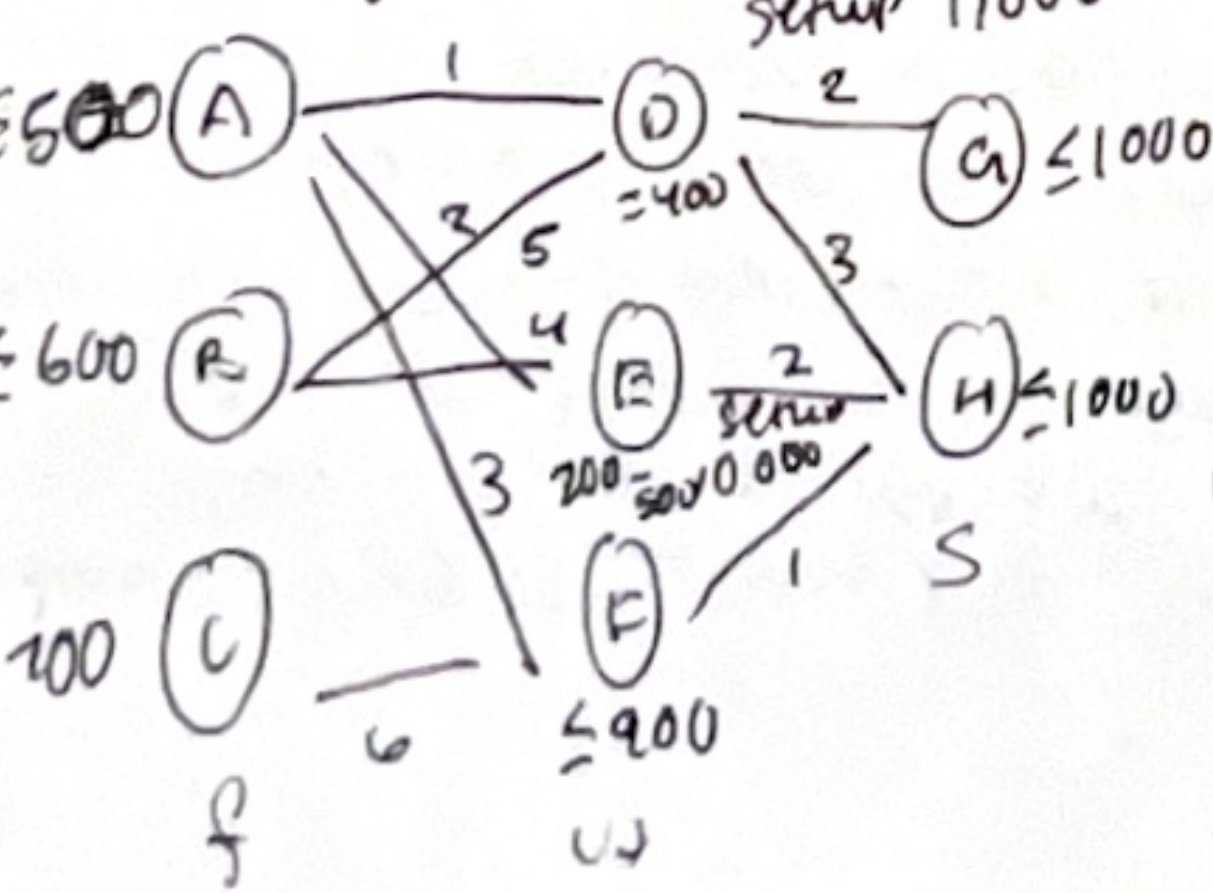
include switched fixed costs in obj. function

add constraints: attach switches to prod. limit

associate actual prod. quant to switched

production limits

multilayer transport and fixed costs example



$x_{AD}, x_{AE}, x_{AF}, x_{BD}, x_{BE}, x_{CE}$ int

$x_{DG}, x_{DH}, x_{EH}, x_{FH}$ int s_D, s_{EH} bin

max rev - fixed - var (F → w) - var (w → s)

$$\max 100x_{DG} + 95(x_{DH} + x_{EH} + x_{FH})$$

$$- (17000s_D + 10000s_{EH})$$

$$- 10(1x_{AD} + 2x_{AE} + 3x_{AF} + 5x_{BD} + 4x_{BE} + 6x_{CE})$$

$$- 10(2x_{DG} + 3x_{DH} + 2x_{EH} + 1x_{FH})$$

fixed constraints

$$x_{AD} + x_{AE} + x_{AF} \leq 500$$

$$x_{BD} + x_{BE} \leq 600$$

$$x_{CE} \leq 700$$

$$x_{AD} + x_{BD} = 400$$

$$x_{AE} + x_{BE} \leq 500$$

$$x_{AF} + x_{BF} \leq 900$$

$$x_{AE} + x_{BE} \geq 200$$

$$x_{AD} + x_{BD} = 400$$

$$x_{DG} + x_{DH} = x_{AD} + x_{BD}$$

$$x_{EH} = x_{AE} + x_{BE}$$

$$x_{FH} = x_{AF} + x_{BF}$$

$$x_{DG} \leq 1000 \quad x_{DH} + x_{EH} + x_{FH} \leq 1000$$

$$x_{DG} \leq (500 + 600)s_D$$

$$x_{EH} \leq (500 + 600)s_{EH}$$

$$x_{EH} \leq (500 + 600)s_{EH}$$

$$x_{EH} \leq (500 + 600)s_{EH}$$

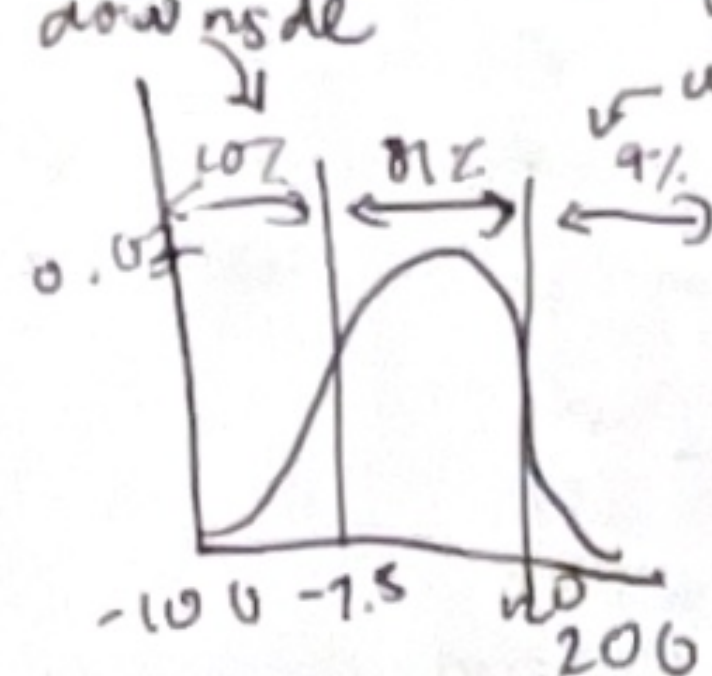
more stuff for exam 2

max downside = max a low percentile ex. max the 10th percentile benefit

on a bad but not abs. worst outcome, how good can u make benefit?

max upside = max a high percentile → best potential good outcome

upside → high end of dist (ex. 90th or 95th percentile)



example prod max & transport

A: 550/unit profit, B: 540/unit profit 100 labor hrs available

2 hrs labor

1 hr labor

Ship from fact: store 1: 54/unit store 2: 56/unit

need at least 30 need at least 20

$$\max 50x_A + 40x_B - 4y_1 - 6y_2 \quad 2x_A + x_B \leq 100 \quad y_1 \geq 30$$

$$y_1 + y_2 \leq x_A + x_B \quad y_2 \geq 20$$